Exponential Smoothing Transcript

Hello everyone, welcome to lesson 2 video 1. In this lesson, we will study more about exponential smoothing. This method gives us weight to recent observations, making forecast more responsive to recent changes in time or in data. In this video, we are going to learn about how do we improve the estimates we have got from moving averages by giving more and more weightage to the recent data.

This is again a very good way of smoothing out fluctuations in the entire time series data by being more focused on what we see around us, near to where we stand. So the key limitation of simple moving average is that it gives equal weightage to all observations. So let's say if we are averaging 3 observations of the entire window, the entire window limit is to average 3 observations to get an estimate, then it is giving each and every observation equal weightages.

While it is not giving more weightage to what we have seen very near, it is giving every data point, whether it is way back in the past or it is very well in the present, it gives equal weightages and that's the pain point or that's the problem with moving averages. So the intuition for the topic we are learning here, which is exponential smoothing, is they do not give equal weightages, but they give some weightage or some more weightage to what the data has seen in the recent values or what the data has seen as recent values in the entire time series. Now this is a simple recursive formula wherein it says that the value at t plus 1 is equals to what the value is, or let's say for example you have alpha which is the smoothing parameter, we call it as a smoothing parameter alpha which is in between 0 to 1, it is dependent on what your current value is and 1 minus alpha times yt minus 1. So let's say it is yt minus 1. So you are giving some weightage to, more weightage to, when you have alpha which is almost closer to 1, you are giving more weightage to more recent value and less weightage to your past value.

Think of it in that way. When you have alpha which is closer to 1, you basically are giving more weightage to this and then less weightage to the second term. That's how it actually works.

Now you see here, the entire simulation we drew for moving averages is what we are trying to draw here. Now with a smaller alpha, how does the entire exponential smoothening looks like and with a higher alpha, how does that look like? Now if you would give more and more preference to nearest data, you actually would see that it replicates the entire series again, right? So it would be giving you same sharp edges. But if you give more weightages to the past, then it will smooth out well, right? And sometimes in some of the use cases, you might want to actually learn from these peaks.

You might not want to eliminate them, which moving average was eliminating because it was taking averages of very high and low values, right? So in this particular case, let's say for example, if you are taking exponential smoothening of these three or these two values, because this is very high, this is low, you have given more weightage to this. Let's say for example, you have given more weightage to this value. In that case, alpha value would be very high.

ence, you see that it has got diminished, but you still see a peak, right? So it also gives you an understanding of what's happening in the real data, okay? And that's what exponential smoothing is all about. So you have a smoothing parameter using which you can actually smoothen out how you would and whether you would like to eliminate the irregularities in the time series or the peaks or fluctuations, right? The sharp edges or do you actually do not want to do that, right? That's what you are able to see here. Now, the limitations of simple exponential smoothing is this is too simple.

The reason for that is the formula that you see does not have any association with trend or seasonality. It doesn't take into consideration whether trend component is involved or seasonal component is involved. But it just takes into consideration what those time series value itself are, right? So because it doesn't take into consideration the trend factor and the seasonality factor, it becomes too simple to work with.

And that's when it also fails to work on or give more realistic or reliable predictions. Now, in the next video, we are going to learn about Holt-Wilters method, which is one of the methods used for seasonality. And this is one of the good methods that people use for predicting time series data or predicting with time series data.

Okay, let's move. Let's meet in the next video then. In summary, exponential smoothing provides a flexible way to forecast by giving more importance to recent data.

It is a widely used business and economics feature, which is very adaptable.

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